SnaanGhar7: A Real-World Bathroom Acoustic Event Dataset for Privacy-Preserving Assistive Living

Debolina Chowdhury[1], Suman Samui[2], Sujoy Saha[1]

[1]Department of Computer Science and Engineering, NIT Durgapur, India

[2]Department of Electronics and Communication Engineering, NIT Durgapur, India

Email: debolinarimuchowdhury@gmail.com,
ssamui.ece@nitdgp.ac.in, sujoy.ju@gmail.com

***Abstract*—Acoustic sensing provides a privacy-preserving alternative to camera-based monitoring for ambient assisted living, yet publicly available datasets for bathroom acoustic event recognition remain scarce. To address this gap, we present *SnaanGhar7*, the first publicly available real-world bathroom acoustic event dataset for intelligent bathroom monitoring. The dataset comprises seven classes—*Flush*, *Shower*, *Bathroom Tap*, *Basin Tap*, *Door*, *Walker/Crutch*, and an *Unknown Class*—collected across five real-world bathroom environments. After manual annotation and sliding-window segmentation, the dataset contains 21,387 near-balanced audio samples with an imbalance ratio of only $1.2\times$. We establish benchmark results using seven representative acoustic event classification models spanning both MFCC- and raw-waveform-based approaches, with the best-performing model achieving 99.64% classification accuracy. By publicly releasing the dataset together with standardized benchmark protocols and baseline implementations, *SnaanGhar7* provides a reproducible benchmark for privacy-preserving bathroom acoustic event recognition, ambient assisted living, and TinyML research.**

Keywords: Bathroom acoustic event recognition; Environmental sound classification; Privacy-preserving sensing; Ambient assisted living; TinyML.

I. Introduction

Acoustic event classification (AEC) has become an important research area in machine learning and signal processing, enabling intelligent systems to recognize human activities and environmental contexts without relying on wearable devices or vision-based sensors. Recent advances in deep learning have significantly improved AEC performance, leading to applications in smart homes, ambient assisted living (AAL), healthcare monitoring, and human–computer interaction [1], [2], [3]. Compared with camera-based systems, acoustic sensing offers a more privacy-preserving solution, making it particularly suitable for monitoring sensitive indoor environments.

Bathrooms are among the most safety-critical areas in residential and assisted-living settings, especially for older adults and individuals with mobility impairments. Monitoring bathroom activities can provide valuable contextual information for healthcare and AAL while preserving user privacy. However, existing environmental sound datasets primarily focus on urban environments, domestic scenes, or general-purpose sound events. Widely used benchmarks such as ESC-50 [4], UrbanSound8K [5], AudioSet [2], FSD50K [6], and TAU Acoustic Scenes [7] contain few or no bathroom-specific recordings, limiting the development and fair evaluation of bathroom acoustic event recognition systems.

To address this gap, we introduce *SnaanGhar7*, the first publicly available multi-site bathroom acoustic event dataset for privacy-preserving bathroom monitoring. The dataset comprises seven representative bathroom sound classes collected from five real-world environments and provides standardized annotations together with benchmark protocols and baseline evaluations using seven representative acoustic event classification models. We believe *SnaanGhar7* will facilitate reproducible research on bathroom acoustic event recognition and privacy-preserving ambient sensing.

The main contributions of this work are summarized as follows:

- We introduce *SnaanGhar7*, the first publicly available multi-site bathroom acoustic event dataset for privacy-preserving acoustic event recognition.
- We present a standardized data collection, annotation, and benchmark generation pipeline for reproducible research.
- We establish baseline results using seven representative deep learning models spanning MFCC- and raw-waveform-based approaches.
- We publicly release the dataset, benchmark protocol, and baseline implementations to support future research in AEC, ambient assisted living, and

TinyML.

II. Related Work

The success of acoustic event classification (AEC) has been largely driven by publicly available benchmark datasets. Widely used datasets, including ESC-50 [4], UrbanSound8K [5], AudioSet [2], FSD50K [6], TAU Acoustic Scenes [7], and DESED [8], have enabled significant advances in environmental sound recognition. However, these datasets primarily target urban, domestic, or general-purpose acoustic events and contain few or no recordings representative of bathroom environments.

Recent research in Ambient Assisted Living (AAL) has increasingly explored privacy-preserving acoustic sensing for monitoring Activities of Daily Living (ADLs). Surveys by Goetze *et al.* [9] and Tong *et al.* [10] highlight the growing interest in audio-based healthcare monitoring while emphasizing the lack of publicly available datasets collected under realistic assisted-living conditions. Existing smart-home datasets, such as the Health Smart Home corpus [11] and the Sweet-Home corpus [12], include only limited bathroom-related recordings, whereas recent bathroom monitoring studies mainly rely on proprietary datasets collected under controlled conditions [13].

Table I compares representative environmental sound datasets and bathroom monitoring studies. Unlike existing resources, *SnaanGhar7* is specifically designed for bathroom acoustic event recognition and provides a publicly available multi-site benchmark with standardized annotations, baseline models, and evaluation protocols.

III. SnaanGhar7: A Bathroom Acoustic Event Dataset

SnaanGhar7 is a publicly available dataset designed for bathroom acoustic event classification and privacy-preserving ambient sensing. It contains seven classes representing common bathroom activities and assistive-living events: *Flush*, *Shower*, *Bathroom Tap*, *Basin Tap*, *Door*, *Walker/Crutch*, and an *Unknown Class*. Approximately 55 minutes of manually annotated recordings were collected for each class across five real-world bathroom environments, resulting in approximately 385 minutes of original audio. The recordings were acquired using a standardized protocol and are intended to support reproducible research in acoustic event classification, ambient assisted living, smart healthcare, and TinyML. Table II summarizes the main characteristics of the dataset.

III-A. Dataset Overview

The dataset consists of six target classes and an *Unknown Class* containing background and non-target sounds, enabling classifiers to reject out-of-distribution acoustic events encountered during real-world deployment.

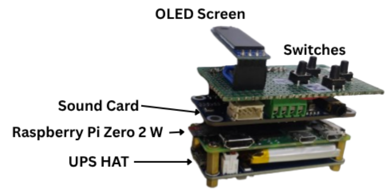

Fig. 1: Embedded recording platform used for data acquisition.

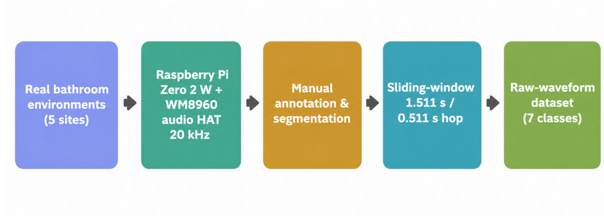

Fig. 2: Overview of the *SnaanGhar7* data collection, annotation, preprocessing, and benchmark generation pipeline.

III-B. Recording Setup

Recordings were collected from five residential, hostel, and institutional bathrooms by three contributors using a custom embedded recording platform based on a Raspberry Pi Zero 2 W and a Waveshare WM8960 Audio HAT (Fig. 1). The different recording environments introduce natural variations in room acoustics, reverberation, plumbing characteristics, and background noise, improving the realism of the dataset.

III-C. Annotation and Preprocessing

Audio recordings were manually annotated and segmented into overlapping analysis windows of 1.51125 s with a hop size of 0.51125 s, corresponding to 30,225 samples at a sampling rate of 20 kHz. These segments constitute the benchmark dataset used throughout this work, resulting in a total of 21,387 near-balanced audio samples. Figure 2 illustrates the complete data collection, annotation, and preprocessing workflow.

IV. Dataset Statistics

The final *SnaanGhar7* corpus contains 21,387 analysis windows with a near-balanced class distribution (Fig. 3). The largest class (*Shower*) contains 3,242 windows, while the smallest (*Door*) contains 2,688 windows, corresponding to a maximum imbalance ratio of only 1.2×. This balanced distribution makes the dataset well suited for benchmarking acoustic event classification models without requiring extensive class re-balancing.

TABLE I: Comparison of representative datasets related to environmental sound recognition and bathroom monitoring.

Dataset	Domain	Bathroom	Public	Benchmark	Audio
ESC-50 [4]	Environmental Sounds	×	✓	✓	✓
UrbanSound8K [5]	Urban Sounds	×	✓	✓	✓
AudioSet [2]	General Audio Events	Partial	✓	✓	✓
FSD50K [6]	General Audio Events	Partial	✓	✓	✓
TAU Acoustic Scenes [7]	Acoustic Scenes	×	✓	✓	✓
Health Smart Home [11]	Smart Home ADLs	Limited	✓	Partial	✓
Sweet-Home [12]	Smart Home ADLs	Limited	✓	Partial	✓
Bathroom Monitoring [9], [14]	Healthcare	✓	×	×	Partial
SnaanGhar7 (Ours)	Bathroom Events	✓	✓	✓	✓

TABLE II: Summary of the *SnaanGhar7* dataset.

Property	Value
Classes	7
Recording environments	5 (Residential, Hostel, Institutional)
Contributors	3
Original duration	≈385 min
Sampling rate	20 kHz
Window / Hop	1.51125 s / 0.51125 s
Total analysis windows	21,387
Class imbalance ratio	1.2×

TABLE III: Class definitions.

Class	Description	Acoustic characteristics
Flush	Toilet flushing	Broadband water flow with transient onset
Shower	Shower water flow	Continuous broadband noise
Bathroom Tap	Bathroom tap running	Continuous water flow with varying intensity
Basin Tap	Basin tap running	Relatively localized continuous water flow
Door	Opening/closing/knocking	Short impulsive sounds
Walker/Crutch	Mobility-aid movement	Repeated transient impacts
Unknown Class	Background/non-target events	Diverse environmental sounds

V. Benchmark Experiments

V-A. Experimental Setup

To establish reproducible baseline results, seven representative acoustic event classification models were evaluated on *SnaanGhar7*, including three MFCC-based models (DS-CNN [15], CRNN [16], and TinyCNN [15]) and four raw-waveform models (ACLNet [17], DPNet [18], ACDNet [19], and SincNet [20]). Audio recordings were segmented into overlapping windows of 1.51125 s with a hop size of 0.51125 s (30,225 samples at 20 kHz). During training, waveform-level amplitude augmentation (±25%) was applied to improve robustness against variations in recording gain and microphone placement. All models were trained using the Adam optimizer with categorical cross-entropy loss. Performance is reported on the held-out test set using overall classification accuracy.

TABLE IV: Recording hardware.

Property	Value
Platform	Raspberry Pi Zero 2 W
Audio interface	Waveshare WM8960 Audio HAT
Sampling rate	20 kHz
Bit depth	32-bit I2S
Channels	Mono

TABLE V: Recording environments.

ID	Location	Type
E01	NIT Durgapur	Institutional
E02	Residential Home, Howrah	Residential
E03	Hall-6, NIT Durgapur	Hostel
E04	DS-1B, NIT Durgapur	Residential
E05	Hall-9, NIT Durgapur	Hostel

V-B. Benchmark Results

Table VII summarizes the benchmark results. All evaluated models achieve accuracies above 93%, with six exceeding 96%. ACDNet achieves the highest accuracy (99.64%), followed closely by TinyCNN (99.63%) and DPNet (99.21%). While ACDNet provides the best offline performance, DPNet offers a favorable balance between recognition accuracy and model complexity, making it suitable for resource-constrained edge deployment.

V-C. Error Analysis

Figure 4 presents the confusion matrix of DPNet [18]. Most classes achieve recognition rates above 98%, indicating strong inter-class separability. The primary confusion occurs between *Flush* and *Bathroom Tap*, which exhibit similar broadband water-flow characteristics, whereas *Basin Tap* is recognized almost perfectly due to its more distinctive acoustic signature.

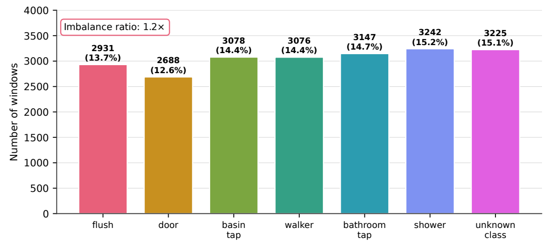

Fig. 3: Class distribution of the *SnaanGhar7* dataset.

TABLE VI: Class-wise distribution of the segmented dataset.

Class	Windows
Flush	2,931
Door	2,688
Basin Tap	3,078
Walker/Crutch	3,076
Bathroom Tap	3,147
Shower	3,242
Unknown Class	3,225
Total	**21,387**

VI. Privacy and Ethical Considerations

SnaanGhar7 is intended for privacy-preserving bathroom activity recognition using passive acoustic sensing rather than cameras. It supports only activity-level acoustic event classification and is not designed for speech recognition, speaker identification, user authentication, or personal attribute inference.

All recordings were collected with informed consent from participants and permission from the respective property owners. A small number of *Unknown Class* recordings contain incidental human speech captured under realistic conditions; these are treated solely as non-target events. No personally identifiable information is included in the released dataset.

Ethics Statement: All participants voluntarily consented to the collection and public release of the anonymized recordings for academic research.

VII. Limitations

SnaanGhar7 contains seven classes collected from five bathroom environments using a single recording platform. Consequently, it does not capture the full diversity of bathroom layouts, devices, or user behaviors. The dataset includes only isolated events and does not provide annotations for overlapping activities. Although the benchmark is intentionally near-balanced, real deployments are typically highly imbalanced. Future versions will expand recording environments, participants, devices, and event categories to improve diversity and generalization.

TABLE VII: Benchmark performance on *SnaanGhar7*.

Model	Input	Params	Acc. (%)	Inf. (ms)
DS-CNN [15]	MFCC	81,863	97.59	39.52
CRNN [16]	MFCC	128,519	98.80	39.13
ACLNet [17]	Raw	146,240	96.82	61.12
DPNet [18]	Raw	547,537	99.21	63.02
TinyCNN [15]	MFCC	1,736,071	99.63	39.00
ACDNet [19]	Raw	4,711,303	**99.64**	66.67
SincNet [20]	Raw	14,149,895	93.76	52.00

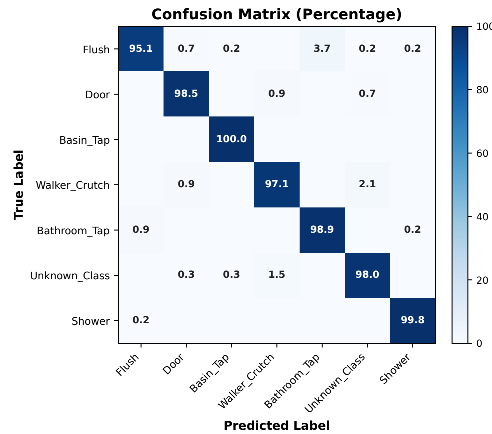

Fig. 4: Confusion matrix of DPNet on *SnaanGhar7*. Most errors occur between acoustically similar water-related events.

VIII. Dataset Availability

The *SnaanGhar7* dataset will be publicly released with audio recordings, annotations, benchmark splits, baseline implementations, and documentation.

Repository: https://github.com/debolina-34/SnaanGhar7
DOI: To be assigned upon publication.
License: CC BY-NC 4.0
Version: v1.0
Contact: `debolinarimuchowdhury@gmail.com`

IX. Conclusion

This paper introduced *SnaanGhar7*, the first publicly available multi-site bathroom acoustic event dataset for privacy-preserving activity recognition. The dataset includes seven classes collected across five real-world environments together with standardized annotations, benchmark protocols, and baseline implementations.

Benchmark results demonstrate that *SnaanGhar7* is both discriminative and suitable for edge deployment, achieving up to 99.64% offline accuracy and 89.26% real-time accuracy on a Raspberry Pi Zero 2 W. We expect *SnaanGhar7* to serve as a reproducible benchmark for acoustic event classification, ambient assisted living, TinyML, and smart healthcare.

References

[1] S. Hershey, S. Chaudhuri, D. P. W. Ellis, J. F. Gemmeke, A. Jansen, R. C. Moore, M. Plakal, D. Platt, R. A. Saurous, B. Seybold, M. Slaney, R. Weiss, and K. Wilson, “Cnn architectures for large-scale audio classification,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 131–135, 2017.

[2] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, “Audio set: An ontology and human-labeled dataset for audio events,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 776–780, 2017.

[3] Y. Tokozume, Y. Ushiku, and T. Harada, “Learning from between-class examples for deep sound recognition,” in *International Conference on Learning Representations (ICLR)*, 2018.

[4] K. J. Piczak, “ESC: Dataset for environmental sound classification,” in *Proc. ACM International Conference on Multimedia*, pp. 1015–1018, 2015.

[5] J. Salamon, C. Jacoby, and J. P. Bello, “A dataset and taxonomy for urban sound research,” in *Proc. ACM International Conference on Multimedia*, pp. 1041–1044, 2014.

[6] E. Fonseca, X. Favory, J. Pons, F. Font, and X. Serra, “Fsd50k: An open dataset of human-labeled sound events,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 829–852, 2022.

[7] A. Mesaros, T. Heittola, and T. Virtanen, “A multi-device dataset for urban acoustic scene classification,” *Proc. Detection and Classification of Acoustic Scenes and Events (DCASE) Workshop*, pp. 9–13, 2018.

[8] R. Serizel, N. Turpault, A. Shah, and J. Salamon, “Sound event detection and separation: A benchmark on the desed dataset,” in *Proceedings of the Detection and Classification of Acoustic Scenes and Events (DCASE) Workshop*, pp. 53–57, 2020.

[9] S. Goetze *et al.*, “Audio-based active and assisted living: A review of selected applications and future trends,” *Computers in Biology and Medicine*, vol. 149, p. 106027, 2022.

[10] K. Tong, K. Attenborough, D. Sharp, S. Taherzadeh, M. Deepak-Gopinath, and J. Vseteckova, “Acceptability of remote monitoring in assisted living/smart homes in the united kingdom and associated use of sounds and vibrations—a systematic review,” *Applied Sciences*, vol. 14, no. 2, p. 843, 2024.

[11] A. Fleury, M. Vacher, and N. Noury, “Svm-based multimodal classification of activities of daily living in health smart homes: Sensors, algorithms, and first experimental results,” in *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, pp. 274–283, 2010.

[12] M. Vacher, B. Lecouteux, P. Chahuara, F. Portet, B. Meillon, and N. Bonnefond, “The sweet-home speech and multimodal corpus for home automation interaction,” in *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC)*, pp. 4499–4506, 2014.

[13] X. Chen *et al.*, “Recognition of bathroom activities in older adults using wearable sensors: A systematic review and recommendations,” *Sensors*, vol. 21, no. 6, p. 2176, 2021.

[14] P. P. Morita, K. S. Sahu, and A. Oetomo, “Health monitoring using smart home technologies: Scoping review,” *JMIR mHealth and uHealth*, vol. 11, p. e37347, 2023.

[15] Y. Zhang, N. Suda, L. Lai, and V. Chandra, “Hello edge: Keyword spotting on microcontrollers,” *arXiv preprint arXiv:1711.07128*, 2017.

[16] E. Cakır, G. Parascandolo, T. Heittola, H. Huttunen, and T. Virtanen, “Convolutional recurrent neural networks for polyphonic sound event detection,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 25, no. 6, pp. 1291–1303, 2017.

[17] J. J. Huang and J. J. A. Leanos, “Aclnet: efficient end-to-end audio classification cnn,” *arXiv preprint arXiv:1811.06669*, 2018.

[18] D. Chowdhury, C. Vinod, S. Samui, M. Saha, and S. Saha, “Dpnet: A lightweight tinyml model for real-time bathroom sound classification,” in *2026 18th International Conference on COMmunication Systems and NETworks (COMSNETS)*, pp. 1409–1414, IEEE, 2026.

[19] M. Mohaimenuzzaman, C. Bergmeir, I. West, and B. Meyer, “Environmental sound classification on the edge: A pipeline for deep acoustic networks on extremely resource-constrained devices,” *Pattern Recognition*, vol. 133, p. 109025, 2023.

[20] M. Ravanelli and Y. Bengio, “Speaker recognition from raw waveform with sincnet,” in *2018 IEEE spoken language technology workshop (SLT)*, pp. 1021–1028, IEEE, 2018.